

lyche-numerical-linear-algebra: residuals

equations measured 2026-09-05T12:57:29Z against the model of 2026-09-01

Corrected (8)

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- E00326 (was)	133	0.118	$\begin{aligned} & \& \boldsymbol{Q}_{\{1\}}:=\left[\right. \\ & \begin{array}{lllll} \boldsymbol{q}_{\{1\}}, & \ldots, & \boldsymbol{q}_{\{n\}} \end{array} \right], & \& \boldsymbol{R}_1 := \\ & \boldsymbol{q}_j := \frac{\boldsymbol{v}_j}{\ \boldsymbol{v}_j\ _2}, \quad j = 1, \dots, n \text{ and} \\ & \left[\begin{array}{ccccc} \ \boldsymbol{v}_1\ _2 & \boldsymbol{a}_2^T \boldsymbol{q}_1 & \boldsymbol{a}_3^T \boldsymbol{q}_1 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_1 \\ 0 & \ \boldsymbol{v}_2\ _2 & \boldsymbol{a}_3^T \boldsymbol{q}_2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_2 \\ & 0 & \ \boldsymbol{v}_3\ _2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_3 \\ & & \ddots & \ddots & \boldsymbol{a}_n^T \boldsymbol{q}_3 \\ & & & \ddots & \vdots \\ & & & \ \boldsymbol{v}_{n-1}\ _2 & \boldsymbol{a}_n^T \boldsymbol{q}_{n-1} \\ & & & 0 & \ \boldsymbol{v}_n\ _2 \end{array} \right]. \end{aligned}$	—	
lyche-numerical-linear-algebra- E00326 (now)	133	—	$\begin{aligned} & \& \boldsymbol{Q}_{\{1\}}:=\left[\right. \\ & \boldsymbol{q}_{\{1\}}, \ldots, \boldsymbol{q}_{\{n\}} \right], \\ & \quad \& \boldsymbol{q}_{\{j\}}:=\frac{\boldsymbol{v}_{\{j\}}}{\left\ \boldsymbol{v}_{\{j\}} \right\ _2}, \quad j=1, \\ & \quad \& \ldots, n \text{ and } \\ & \& \boldsymbol{R}_{\{1\}}:=\left[\right. \\ & \begin{array}{cccccc} \left\ \boldsymbol{v}_{\{1\}} \right\ _2 & \boldsymbol{a}_{\{2\}}^T \boldsymbol{q}_{\{1\}} & \boldsymbol{a}_{\{3\}}^T \boldsymbol{q}_{\{1\}} & \cdots & \boldsymbol{a}_{\{n-1\}}^T \boldsymbol{q}_{\{1\}} \\ 0 & \left\ \boldsymbol{v}_{\{2\}} \right\ _2 & \boldsymbol{a}_{\{3\}}^T \boldsymbol{q}_{\{2\}} & \cdots & \boldsymbol{a}_{\{n-1\}}^T \boldsymbol{q}_{\{2\}} \\ & 0 & \left\ \boldsymbol{v}_{\{3\}} \right\ _2 & \cdots & \boldsymbol{a}_{\{n-1\}}^T \boldsymbol{q}_{\{3\}} \\ & & \ddots & \ddots & \vdots \\ & & & \ddots & \left\ \boldsymbol{v}_{\{n-1\}} \right\ _2 \\ & & & & 0 \end{array} \\ & \boldsymbol{a}_{\{n\}}^T \boldsymbol{q}_{\{n-1\}} \\ & \left. \left\ \boldsymbol{v}_{\{n\}} \right\ _2 \right]. \end{aligned}$	—	
basis: inferred / ink					
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value					
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- EQ0369 (was)	150	■0.298	$\boldsymbol{J}:=\operatorname{diag}\left(\boldsymbol{U}_1, \boldsymbol{U}_2\right)=\left[\begin{array}{cccccc} 2 & 1 & 0 & & & \\ 0 & 2 & 1 & & & \\ 0 & 0 & 2 & & 1 & \\ & & 0 & 1 & & \\ & & 0 & 2 & & \\ & & & & 0 & \\ & & & & 0 & 1 \end{array}\right] \in \mathbb{R}^{8 \times 8} .$	—	
lyche-numerical-linear-algebra- EQ0369 (now)	150	—	$d_i \text { for } i=1, \ldots, n, b_{i+1, i}:=a_i, b_{i+1, i}:=c_i \text { for } i \text { otherwise. } \in \mathbb{C}^{m \times n} \text { and } 1 \leq i_1 < i_2 < \cdots < i_r \leq m, 1 \leq j \text { matrix } A\left(i, j\right) \in \mathbb{C}^{r \times c} \text { is the submatrix of } A$ $:=c_i \text { for } i \text { \textit{otherwise.}} \in \mathbb{C}^{m \times n}$ $n\} \text { and } 1 \leq i_1 < i_2 < \cdots < i_r \leq m, 1 \leq j$ $\text { \textit{matrix} } A\left(\boldsymbol{i}, \boldsymbol{j}\right) \in \mathbb{C}^{r \times c} \text { is the submatrix of } A$ $\text { is the submatrix of } A$	$d_i \text { for } i=1, \ldots, n, b_{i+1, i}:=a_i, b_{i+1, i}:=c_i \text { for } i \text { otherwise. } \in \mathbb{C}^{m \times n} \text { and } 1 \leq i_1 < i_2 < \cdots < i_r \leq m, 1 \leq j \text { matrix } A\left(i, j\right) \in \mathbb{C}^{r \times c} \text { is the submatrix of } A$	—
basis: inferred / ink					
lyche-numerical-linear-algebra- EQ0579 (was)	215	■0.322	$\begin{array}{l} \boldsymbol{A}^* \boldsymbol{A} \\ \boldsymbol{x} & =\left[\begin{array}{ccc} 1 & \cdots & 1 \\ t_1 & \cdots & t_m \end{array}\right] \left[\begin{array}{c} 1 \\ \vdots \\ 1 \\ 1 \\ m \end{array}\right] \left[\begin{array}{c} x_1 \\ x_2 \end{array}\right]=\left[\begin{array}{cc} m & \sum t_k \\ \sum t_k & \sum t_k^2 \end{array}\right] \left[\begin{array}{c} x_1 \\ x_2 \end{array}\right], \\ \left[\begin{array}{ccc} 1 & \cdots & 1 \\ t_1 & \cdots & t_m \end{array}\right] \left[\begin{array}{c} y_1 \\ \vdots \\ y_m \end{array}\right]=\left[\begin{array}{c} \sum y_k \\ \sum t_k y_k \end{array}\right]=\boldsymbol{A}^* \boldsymbol{b}, \end{array}$	$\boldsymbol{A}^* \boldsymbol{A} x=\left[\begin{array}{ccc} 1 & \cdots & 1 \\ t_1 & \cdots & t_m \end{array}\right] \left[\begin{array}{c} 1 \\ \vdots \\ 1 \\ 1 \\ m \end{array}\right] \left[\begin{array}{c} x_1 \\ x_2 \end{array}\right]=\left[\begin{array}{cc} m & \sum t_k \\ \sum t_k & \sum t_k^2 \end{array}\right] \left[\begin{array}{c} x_1 \\ x_2 \end{array}\right],$ $=\left[\begin{array}{ccc} 1 & \cdots & 1 \\ t_1 & \cdots & t_m \end{array}\right] \left[\begin{array}{c} y_1 \\ \vdots \\ y_m \end{array}\right]=\left[\begin{array}{c} \sum y_k \\ \sum t_k y_k \end{array}\right]=\boldsymbol{A}^* \boldsymbol{b},$	—
lyche-numerical-linear-algebra- EQ0579 (now)	215	—	$\text {\textit{nal} if } a_{i j}=0 \text {\textit{for} } i \neq j . \\ \text {\textit{triangular or \textbf{right triangular} if } } \\ a_{i j}=0 \text {\textit{for} } i > j . \\ \text {\textit{triangular or \textbf{left triangular} if } } a_{i j}=0 \text {\textit{for} } i < j . \\ \text {\textit{Hessenberg if } } a_{i j}=0 \text {\textit{for} } i > j+1 . \\ \text {\textit{Hessenberg if } } a_{i j}=0 \text {\textit{for} } i < j+1 . \\ \text {\textit{gonal if } } a_{i j}=0 \text {\textit{for} } i-j > 1 . \\ \text {\textit{ded if } } a_{i j}=0 \text {\textit{for} } i-j > d . \\ \text {\textit{the following notations for diagonal- and tridiagonal } } n \times n$	$(not\ rendered)$	—
basis: inferred / ink					
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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lyche-numerical-linear-algebra- E00729 (now)	258	—	$(DX + XD)_{j,k} = \sum_{\ell=1}^m d_{j,\ell} x_{\ell,k} + \sum_{\ell=1}^m x_{j,\ell} d_{\ell,k} = \lambda_j x_{j,k} + \lambda_k x_{j,k}$ for all j, k	$(DX + XD)_{j,k} = \sum_{\ell=1}^m d_{j,\ell} x_{\ell,k} + \sum_{\ell=1}^m x_{j,\ell} d_{\ell,k} = \lambda_j x_{j,k} + \lambda_k x_{j,k}$ for all j, k	—
basis: inferred / ink					
lyche-numerical-linear-algebra- E00763 (was)	271	■ 0.402	$\boldsymbol{G}_{\omega} = \begin{bmatrix} l & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} \omega/2 & 0 \\ \omega^2/4 & \omega/2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 1-\omega & \omega/2 \\ \omega(1-\omega)/2 & 1-\omega+\omega^2/4 \end{bmatrix}.$	$G_{\omega} = \begin{bmatrix} l & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} \omega/2 & 0 \\ \omega^2/4 & \omega/2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 1-\omega & \omega/2 \\ \omega(1-\omega)/2 & 1-\omega+\omega^2/4 \end{bmatrix}.$	—
lyche-numerical-linear-algebra- E00763 (now)	271	—	$\langle j_1, j_2 \mid j_c \rangle \begin{bmatrix} \cdot & \cdot & \cdots & \cdot \\ a_{i_r, j_1} & a_{i_r, j_2} & \cdots & a_{i_r, j_c} \end{bmatrix}$	$\langle j_1, j_2 \mid j_c \rangle \begin{bmatrix} \cdot & \cdot & \cdots & \cdot \\ a_{i_r, j_1} & a_{i_r, j_2} & \cdots & a_{i_r, j_c} \end{bmatrix}$	—
basis: inferred / ink					
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Unresolved

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-EQ0855	296	■0.386 ● C -60	<pre>\begin{array}{ll} \boldsymbol{t}_0=\left[\begin{array}{c} 3 \\ -3/2 \end{array}\right], \quad \boldsymbol{x}_1=\left[\begin{array}{c} -1/4 \\ -1/2 \end{array}\right], \\ \boldsymbol{p}_1=\left[\begin{array}{c} 3/8 \\ 3/4 \end{array}\right], \quad \boldsymbol{t}_1=\left[\begin{array}{c} 0 \\ 9/8 \end{array}\right], \quad \alpha_1=2/3, \quad \boldsymbol{x}_2=\mathbf{0}, \quad \boldsymbol{r}_2=\mathbf{0}. \end{array}</pre>	$\boldsymbol{t}_0 = \begin{bmatrix} 3 \\ -3/2 \end{bmatrix}, \quad \alpha_0 = 1/2, \quad \boldsymbol{x}_1 = \begin{bmatrix} -1/4 \\ -1/2 \end{bmatrix}, \quad \boldsymbol{r}_1 = \begin{bmatrix} 0 \\ 3/4 \end{bmatrix}, \quad \beta_0 = 1/4,$ $\boldsymbol{p}_1 = \begin{bmatrix} 3/8 \\ 3/4 \end{bmatrix}, \quad \boldsymbol{t}_1 = \begin{bmatrix} 0 \\ 9/8 \end{bmatrix}, \quad \alpha_1 = 2/3, \quad \boldsymbol{x}_2 = \mathbf{0}, \quad \boldsymbol{r}_2 = \mathbf{0}.$	
lyche-numerical-linear-algebra-EQ1043	353	■0.971 ● C +78	<pre>\begin{figure} \[\boldsymbol{A}=\left[\begin{array}{lllll} x & x & x & x & x \\ x & x & x & x & x \\ x & x & x & x & x \\ x & x & x & x & x \\ x & x & x & x & x \end{array}\right] \xrightarrow{\boldsymbol{P}_{12}^*} \left[\begin{array}{lllll} x & x & x & x & x \\ \mathbf{x} & x & x & x & x \\ 0 & 0 & x & x & x \\ 0 & 0 & 0 & x & x \end{array}\right] \xrightarrow{\boldsymbol{P}_{23}^*} \left[\begin{array}{lllll} x & x & x & x & x \\ x & x & x & x & x \\ 0 & \mathbf{x} & x & x & x \\ 0 & 0 & 0 & x & x \end{array}\right] \xrightarrow{\boldsymbol{P}_{34}^*} \left[\begin{array}{lllll} x & x & x & x & x \\ x & x & x & x & x \\ 0 & x & x & x & x \\ 0 & 0 & \mathbf{x} & x & x \end{array}\right]. \]</pre>	(not rendered)	$\boldsymbol{A} = \begin{bmatrix} x & x & x & x & x \\ 0 & x & x & x & x \\ 0 & 0 & x & x & x \\ 0 & 0 & 0 & x & x \end{bmatrix} \xrightarrow{\boldsymbol{P}_{12}^*} \begin{bmatrix} x & x & x & x & x \\ \mathbf{x} & x & x & x & x \\ 0 & 0 & x & x & x \\ 0 & 0 & 0 & x & x \end{bmatrix} \xrightarrow{\boldsymbol{P}_{23}^*} \begin{bmatrix} x & x & x & x & x \\ x & x & x & x & x \\ 0 & \mathbf{x} & x & x & x \\ 0 & 0 & 0 & x & x \end{bmatrix} \xrightarrow{\boldsymbol{P}_{34}^*} \begin{bmatrix} x & x & x & x & x \\ x & x & x & x & x \\ 0 & x & x & x & x \\ 0 & 0 & \mathbf{x} & x & x \end{bmatrix}.$
lyche-numerical-linear-algebra-EQ0195 (3.9)	82	■1.000 ● C +10	<pre>\begin{figure} \captionsetup{labelformat=empty} \caption{Listing 3.3 cforwardsolve} \[N_{LU}:=\frac{2}{3} n^3-\frac{1}{2} n^2-\frac{1}{6} n \]</pre>	(not rendered)	$N_{LU} := \frac{2}{3}n^3 - \frac{1}{2}n^2 - \frac{1}{6}n$
lyche-numerical-linear-algebra-EQ0232 (3.22)	93	■1.000 ● C +54	<pre>\begin{figure} \captionsetup{labelformat=empty} \caption{Listing 3.5 cbacksolve} \[\boldsymbol{A}((k+1):n,(k+1):n) \boldsymbol{b}_{\{k\}((k+1):n)}=-\boldsymbol{A}((k+1):n,k) \boldsymbol{b}_{\{k\}(k)}, \quad k=1, \ldots, n-1. \]</pre>	(not rendered)	$\boldsymbol{A}((k+1):n,(k+1):n)\boldsymbol{b}_k((k+1):n)=-\boldsymbol{A}((k+1):n,k)\boldsymbol{b}_k(k), \quad k=1,\ldots,n-1.$
lyche-numerical-linear-algebra-EQ0258 (4.7)	106	■1.000 ● C +54	<pre>\begin{figure} \captionsetup{labelformat=empty} \caption{Listing 4.2 bandcholesky} \[0<(\leq)\left(\alpha \boldsymbol{e}_i+\beta \boldsymbol{e}_j\right)^* \boldsymbol{A}\left(\alpha \boldsymbol{e}_i+\beta \boldsymbol{e}_j\right)=\left \alpha\right ^2 a_{i i}+\left \beta\right ^2 a_{j j}+2 \operatorname{Re}\left(\bar{\alpha} \beta a_{i j}\right).</pre>	(not rendered)	$0 < (\leq)(\alpha \boldsymbol{e}_i + \beta \boldsymbol{e}_j)^* \boldsymbol{A}(\alpha \boldsymbol{e}_i + \beta \boldsymbol{e}_j) = \alpha ^2 a_{ii} + \beta ^2 a_{jj} + 2 \operatorname{Re}(\bar{\alpha} \beta a_{ij}).$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Flagged, not acted on

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-E00485	191	0.917 C +32	$\begin{aligned} & \ \boldsymbol{A}\boldsymbol{B}\ = \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{B}\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{B}\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{B}\boldsymbol{x}\ } \frac{\ \boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } \\ & \leq \max_{\boldsymbol{y} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{y}\ }{\ \boldsymbol{y}\ } \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \ \boldsymbol{A}\ \ \boldsymbol{B}\ \end{aligned}$	$\begin{aligned} \ \boldsymbol{A}\boldsymbol{B}\ &= \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{B}\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{B}\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{B}\boldsymbol{x}\ } \frac{\ \boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } \\ &\leq \max_{\boldsymbol{y} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{y}\ }{\ \boldsymbol{y}\ } \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \ \boldsymbol{A}\ \ \boldsymbol{B}\ . \end{aligned}$	
Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

194 more flagged rows below the band, stated as a count: 30 component, 164 weak; confidence high 138, mid 35, low 21, none 0